In [1]:
```
import pandas as pd
import numpy as np
```

In [3]:
```
titanic = pd.read_csv(r"C:\Users\LIKHITH\OneDrive\Documents\titanic dataset.csv", h
```

In [4]:
```
titanic
```

Out[4]:

| | PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | F |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | Braund, Mr. Owen Harris | male | 22.0 | 1 | 0 | A/5 21171 | 7.2 |
| **1** | 2 | 1 | 1 | Cumings, Mrs. John Bradley (Florence Briggs Th... | female | 38.0 | 1 | 0 | PC 17599 | 71.2 |
| **2** | 3 | 1 | 3 | Heikkinen, Miss. Laina | female | 26.0 | 0 | 0 | STON/O2. 3101282 | 7.9 |
| **3** | 4 | 1 | 1 | Futrelle, Mrs. Jacques Heath (Lily May Peel) | female | 35.0 | 1 | 0 | 113803 | 53.1 |
| **4** | 5 | 0 | 3 | Allen, Mr. William Henry | male | 35.0 | 0 | 0 | 373450 | 8.0 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | |
| **886** | 887 | 0 | 2 | Montvila, Rev. Juozas | male | 27.0 | 0 | 0 | 211536 | 13.0 |
| **887** | 888 | 1 | 1 | Graham, Miss. Margaret Edith | female | 19.0 | 0 | 0 | 112053 | 30.0 |
| **888** | 889 | 0 | 3 | Johnston, Miss. Catherine Helen "Carrie" | female | NaN | 1 | 2 | W./C. 6607 | 23.4 |
| **889** | 890 | 1 | 1 | Behr, Mr. Karl Howell | male | 26.0 | 0 | 0 | 111369 | 30.0 |
| **890** | 891 | 0 | 3 | Dooley, Mr. Patrick | male | 32.0 | 0 | 0 | 370376 | 7.7 |

891 rows × 12 columns

In [5]:
```
titanic.tail()
```

Out[5]:

| | PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare |
|---|---|---|---|---|---|---|---|---|---|---|
| **886** | 887 | 0 | 2 | Montvila, Rev. Juozas | male | 27.0 | 0 | 0 | 211536 | 13.00 |
| **887** | 888 | 1 | 1 | Graham, Miss. Margaret Edith | female | 19.0 | 0 | 0 | 112053 | 30.00 |
| **888** | 889 | 0 | 3 | Johnston, Miss. Catherine Helen "Carrie" | female | NaN | 1 | 2 | W./C. 6607 | 23.45 |
| **889** | 890 | 1 | 1 | Behr, Mr. Karl Howell | male | 26.0 | 0 | 0 | 111369 | 30.00 |
| **890** | 891 | 0 | 3 | Dooley, Mr. Patrick | male | 32.0 | 0 | 0 | 370376 | 7.75 |

In [6]:
```
titanic.head()
```

Out[6]:

| | PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | Braund, Mr. Owen Harris | male | 22.0 | 1 | 0 | A/5 21171 | 7.2500 |
| **1** | 2 | 1 | 1 | Cumings, Mrs. John Bradley (Florence Briggs Th... | female | 38.0 | 1 | 0 | PC 17599 | 71.2833 |
| **2** | 3 | 1 | 3 | Heikkinen, Miss. Laina | female | 26.0 | 0 | 0 | STON/O2. 3101282 | 7.9250 |
| **3** | 4 | 1 | 1 | Futrelle, Mrs. Jacques Heath (Lily May Peel) | female | 35.0 | 1 | 0 | 113803 | 53.1000 |
| **4** | 5 | 0 | 3 | Allen, Mr. William Henry | male | 35.0 | 0 | 0 | 373450 | 8.0500 |

In [7]:

```
titanic.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [8]:

```
titanic.describe()
```

Out[8]:

| | PassengerId | Survived | Pclass | Age | SibSp | Parch | Fare |
|---|---|---|---|---|---|---|---|
| **count** | 891.000000 | 891.000000 | 891.000000 | 714.000000 | 891.000000 | 891.000000 | 891.00000 |
| **mean** | 446.000000 | 0.383838 | 2.308642 | 29.699118 | 0.523008 | 0.381594 | 32.20420 |
| **std** | 257.353842 | 0.486592 | 0.836071 | 14.526497 | 1.102743 | 0.806057 | 49.69342 |
| **min** | 1.000000 | 0.000000 | 1.000000 | 0.420000 | 0.000000 | 0.000000 | 0.00000 |
| **25%** | 223.500000 | 0.000000 | 2.000000 | 20.125000 | 0.000000 | 0.000000 | 7.91040 |
| **50%** | 446.000000 | 0.000000 | 3.000000 | 28.000000 | 0.000000 | 0.000000 | 14.45420 |
| **75%** | 668.500000 | 1.000000 | 3.000000 | 38.000000 | 1.000000 | 0.000000 | 31.00000 |
| **max** | 891.000000 | 1.000000 | 3.000000 | 80.000000 | 8.000000 | 6.000000 | 512.32920 |

In [9]:
```
#Name column can never decide survival of a person, hence we can safely delete it
del titanic["Name"]
titanic.head()
```

Out[9]:

| | PassengerId | Survived | Pclass | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | E |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | male | 22.0 | 1 | 0 | A/5 21171 | 7.2500 | NaN | |
| **1** | 2 | 1 | 1 | female | 38.0 | 1 | 0 | PC 17599 | 71.2833 | C85 | |
| **2** | 3 | 1 | 3 | female | 26.0 | 0 | 0 | STON/O2. 3101282 | 7.9250 | NaN | |
| **3** | 4 | 1 | 1 | female | 35.0 | 1 | 0 | 113803 | 53.1000 | C123 | |
| **4** | 5 | 0 | 3 | male | 35.0 | 0 | 0 | 373450 | 8.0500 | NaN | |

In [10]:
```
del titanic["Ticket"]
titanic.head()
```

Out[10]:

| | PassengerId | Survived | Pclass | Sex | Age | SibSp | Parch | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | male | 22.0 | 1 | 0 | 7.2500 | NaN | S |
| **1** | 2 | 1 | 1 | female | 38.0 | 1 | 0 | 71.2833 | C85 | C |
| **2** | 3 | 1 | 3 | female | 26.0 | 0 | 0 | 7.9250 | NaN | S |
| **3** | 4 | 1 | 1 | female | 35.0 | 1 | 0 | 53.1000 | C123 | S |
| **4** | 5 | 0 | 3 | male | 35.0 | 0 | 0 | 8.0500 | NaN | S |

In [11]:
```
del titanic["Fare"]
titanic.head()
```

Out[11]:

| | PassengerId | Survived | Pclass | Sex | Age | SibSp | Parch | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | male | 22.0 | 1 | 0 | NaN | S |
| **1** | 2 | 1 | 1 | female | 38.0 | 1 | 0 | C85 | C |
| **2** | 3 | 1 | 3 | female | 26.0 | 0 | 0 | NaN | S |
| **3** | 4 | 1 | 1 | female | 35.0 | 1 | 0 | C123 | S |
| **4** | 5 | 0 | 3 | male | 35.0 | 0 | 0 | NaN | S |

In [12]:

```
del titanic['Cabin']
titanic.head()
```

Out[12]:

| | PassengerId | Survived | Pclass | Sex | Age | SibSp | Parch | Embarked |
|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | male | 22.0 | 1 | 0 | S |
| **1** | 2 | 1 | 1 | female | 38.0 | 1 | 0 | C |
| **2** | 3 | 1 | 3 | female | 26.0 | 0 | 0 | S |
| **3** | 4 | 1 | 1 | female | 35.0 | 1 | 0 | S |
| **4** | 5 | 0 | 3 | male | 35.0 | 0 | 0 | S |

In [13]:

```
# Changing Value for "Male, Female" string values to numeric values , male=1 and fe
def getNumber(str):
    if str=="male":
        return 1
    else:
        return 2
titanic["Gender"]=titanic["Sex"].apply(getNumber)
#We have created a new column called "Gender" and
#filling it with values 1,2 based on the values of sex column
titanic.head()
```

Out[13]:

| | PassengerId | Survived | Pclass | Sex | Age | SibSp | Parch | Embarked | Gender |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | male | 22.0 | 1 | 0 | S | 1 |
| **1** | 2 | 1 | 1 | female | 38.0 | 1 | 0 | C | 2 |
| **2** | 3 | 1 | 3 | female | 26.0 | 0 | 0 | S | 2 |
| **3** | 4 | 1 | 1 | female | 35.0 | 1 | 0 | S | 2 |
| **4** | 5 | 0 | 3 | male | 35.0 | 0 | 0 | S | 1 |

In [14]:

```
#Deleting Sex column, since no use of it now
del titanic["Sex"]
titanic.head()
```

Out[14]:

| | PassengerId | Survived | Pclass | Age | SibSp | Parch | Embarked | Gender |
|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 22.0 | 1 | 0 | S | 1 |
| **1** | 2 | 1 | 1 | 38.0 | 1 | 0 | C | 2 |
| **2** | 3 | 1 | 3 | 26.0 | 0 | 0 | S | 2 |
| **3** | 4 | 1 | 1 | 35.0 | 1 | 0 | S | 2 |
| **4** | 5 | 0 | 3 | 35.0 | 0 | 0 | S | 1 |

In [15]:

```
titanic.isnull().sum()
```

Out[15]:

```
PassengerId      0
Survived         0
Pclass           0
Age            177
SibSp            0
Parch            0
Embarked         2
Gender           0
dtype: int64
```

In [16]:

```
meanS= titanic[titanic.Survived==1].Age.mean()
meanS
```

Out[16]:

```
np.float64(28.343689655172415)
```

In [17]:

```
titanic["age"]=np.where(pd.isnull(titanic.Age) & titanic["Survived"]==1  ,meanS, ti
titanic.head()
```

Out[17]:

| | PassengerId | Survived | Pclass | Age | SibSp | Parch | Embarked | Gender | age |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 22.0 | 1 | 0 | S | 1 | 22.0 |
| **1** | 2 | 1 | 1 | 38.0 | 1 | 0 | C | 2 | 38.0 |
| **2** | 3 | 1 | 3 | 26.0 | 0 | 0 | S | 2 | 26.0 |
| **3** | 4 | 1 | 1 | 35.0 | 1 | 0 | S | 2 | 35.0 |
| **4** | 5 | 0 | 3 | 35.0 | 0 | 0 | S | 1 | 35.0 |

In [18]:

```
titanic
```

Out[18]:

| | PassengerId | Survived | Pclass | Age | SibSp | Parch | Embarked | Gender | age |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 22.0 | 1 | 0 | S | 1 | 22.0 |
| **1** | 2 | 1 | 1 | 38.0 | 1 | 0 | C | 2 | 38.0 |
| **2** | 3 | 1 | 3 | 26.0 | 0 | 0 | S | 2 | 26.0 |
| **3** | 4 | 1 | 1 | 35.0 | 1 | 0 | S | 2 | 35.0 |
| **4** | 5 | 0 | 3 | 35.0 | 0 | 0 | S | 1 | 35.0 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **886** | 887 | 0 | 2 | 27.0 | 0 | 0 | S | 1 | 27.0 |
| **887** | 888 | 1 | 1 | 19.0 | 0 | 0 | S | 2 | 19.0 |
| **888** | 889 | 0 | 3 | NaN | 1 | 2 | S | 2 | NaN |
| **889** | 890 | 1 | 1 | 26.0 | 0 | 0 | C | 1 | 26.0 |
| **890** | 891 | 0 | 3 | 32.0 | 0 | 0 | Q | 1 | 32.0 |

891 rows × 9 columns

In [19]:

```
titanic.isnull().sum()
```

Out[19]:

```
PassengerId      0
Survived         0
Pclass           0
Age            177
SibSp            0
Parch            0
Embarked         2
Gender           0
age            125
dtype: int64
```

In [20]:

```
# Finding the mean age of "Not Survived" people
meanNS=titanic[titanic.Survived==0].Age.mean()
meanNS
```

Out[20]:

```
np.float64(30.62617924528302)
```

In [21]:

```
titanic.age.fillna(meanNS,inplace=True)
titanic.head()
```

```
C:\Users\LIKHITH\AppData\Local\Temp\ipykernel_18756\1157731433.py:1: FutureWarning:
A value is trying to be set on a copy of a DataFrame or Series through chained assig
nment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because
the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method
({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform
the operation inplace on the original object.


  titanic.age.fillna(meanNS,inplace=True)
```

Out[21]:

| | PassengerId | Survived | Pclass | Age | SibSp | Parch | Embarked | Gender | age |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 22.0 | 1 | 0 | S | 1 | 22.0 |
| **1** | 2 | 1 | 1 | 38.0 | 1 | 0 | C | 2 | 38.0 |
| **2** | 3 | 1 | 3 | 26.0 | 0 | 0 | S | 2 | 26.0 |
| **3** | 4 | 1 | 1 | 35.0 | 1 | 0 | S | 2 | 35.0 |
| **4** | 5 | 0 | 3 | 35.0 | 0 | 0 | S | 1 | 35.0 |

In [22]:
```
titanic
```

Out[22]:

| | PassengerId | Survived | Pclass | Age | SibSp | Parch | Embarked | Gender | age |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 22.0 | 1 | 0 | S | 1 | 22.000000 |
| **1** | 2 | 1 | 1 | 38.0 | 1 | 0 | C | 2 | 38.000000 |
| **2** | 3 | 1 | 3 | 26.0 | 0 | 0 | S | 2 | 26.000000 |
| **3** | 4 | 1 | 1 | 35.0 | 1 | 0 | S | 2 | 35.000000 |
| **4** | 5 | 0 | 3 | 35.0 | 0 | 0 | S | 1 | 35.000000 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **886** | 887 | 0 | 2 | 27.0 | 0 | 0 | S | 1 | 27.000000 |
| **887** | 888 | 1 | 1 | 19.0 | 0 | 0 | S | 2 | 19.000000 |
| **888** | 889 | 0 | 3 | NaN | 1 | 2 | S | 2 | 30.626179 |
| **889** | 890 | 1 | 1 | 26.0 | 0 | 0 | C | 1 | 26.000000 |
| **890** | 891 | 0 | 3 | 32.0 | 0 | 0 | Q | 1 | 32.000000 |

891 rows × 9 columns

In [23]:
```
titanic.isnull().sum()
```

Out[23]:
```
PassengerId      0
Survived         0
Pclass           0
Age            177
SibSp            0
Parch            0
Embarked         2
Gender           0
age              0
dtype: int64
```

In [24]:
```
del titanic['Age']
titanic.head()
```

Out[24]:

| | PassengerId | Survived | Pclass | SibSp | Parch | Embarked | Gender | age |
|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 1 | 0 | S | 1 | 22.0 |
| **1** | 2 | 1 | 1 | 1 | 0 | C | 2 | 38.0 |
| **2** | 3 | 1 | 3 | 0 | 0 | S | 2 | 26.0 |
| **3** | 4 | 1 | 1 | 1 | 0 | S | 2 | 35.0 |
| **4** | 5 | 0 | 3 | 0 | 0 | S | 1 | 35.0 |

In [25]:
```
# Finding the number of people who have survived
# given that they have embarked or boarded from a particular port

survivedQ = titanic[titanic.Embarked == 'Q'][titanic.Survived == 1].shape[0]
survivedC = titanic[titanic.Embarked == 'C'][titanic.Survived == 1].shape[0]
survivedS = titanic[titanic.Embarked == 'S'][titanic.Survived == 1].shape[0]
print(survivedQ)
print(survivedC)
print(survivedS)
```

```
30
93
217
```

```
C:\Users\LIKHITH\AppData\Local\Temp\ipykernel_18756\3300902897.py:4: UserWarning: Bo
olean Series key will be reindexed to match DataFrame index.
  survivedQ = titanic[titanic.Embarked == 'Q'][titanic.Survived == 1].shape[0]
C:\Users\LIKHITH\AppData\Local\Temp\ipykernel_18756\3300902897.py:5: UserWarning: Bo
olean Series key will be reindexed to match DataFrame index.
  survivedC = titanic[titanic.Embarked == 'C'][titanic.Survived == 1].shape[0]
C:\Users\LIKHITH\AppData\Local\Temp\ipykernel_18756\3300902897.py:6: UserWarning: Bo
olean Series key will be reindexed to match DataFrame index.
  survivedS = titanic[titanic.Embarked == 'S'][titanic.Survived == 1].shape[0]
```

In [26]:
```
survivedQ = titanic[titanic.Embarked == 'Q'][titanic.Survived == 0].shape[0]
survivedC = titanic[titanic.Embarked == 'C'][titanic.Survived == 0].shape[0]
survivedS = titanic[titanic.Embarked == 'S'][titanic.Survived == 0].shape[0]
print(survivedQ)
print(survivedC)
print(survivedS)
```

```
47
75
427
C:\Users\LIKHITH\AppData\Local\Temp\ipykernel_18756\3240960939.py:1: UserWarning: Bo
olean Series key will be reindexed to match DataFrame index.
  survivedQ = titanic[titanic.Embarked == 'Q'][titanic.Survived == 0].shape[0]
C:\Users\LIKHITH\AppData\Local\Temp\ipykernel_18756\3240960939.py:2: UserWarning: Bo
olean Series key will be reindexed to match DataFrame index.
  survivedC = titanic[titanic.Embarked == 'C'][titanic.Survived == 0].shape[0]
C:\Users\LIKHITH\AppData\Local\Temp\ipykernel_18756\3240960939.py:3: UserWarning: Bo
olean Series key will be reindexed to match DataFrame index.
  survivedS = titanic[titanic.Embarked == 'S'][titanic.Survived == 0].shape[0]
```

In [27]:
```
titanic.dropna(inplace=True)
titanic.head()
```

Out[27]:

| | PassengerId | Survived | Pclass | SibSp | Parch | Embarked | Gender | age |
|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 1 | 0 | S | 1 | 22.0 |
| **1** | 2 | 1 | 1 | 1 | 0 | C | 2 | 38.0 |
| **2** | 3 | 1 | 3 | 0 | 0 | S | 2 | 26.0 |
| **3** | 4 | 1 | 1 | 1 | 0 | S | 2 | 35.0 |
| **4** | 5 | 0 | 3 | 0 | 0 | S | 1 | 35.0 |

In [32]:
```
titanic.isnull().sum()
```

Out[32]:
```
PassengerId    0
Survived       0
Pclass         0
SibSp          0
Parch          0
Embarked       0
Sex            0
Age            0
Embark         0
dtype: int64
```

In [34]:
```
titanic.head()
```

Out[34]:

| | PassengerId | Survived | Pclass | SibSp | Parch | Embarked | Sex | Age | Embark |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 1 | 0 | S | 1 | 22.0 | 1 |
| **1** | 2 | 1 | 1 | 1 | 0 | C | 2 | 38.0 | 3 |
| **2** | 3 | 1 | 3 | 0 | 0 | S | 2 | 26.0 | 1 |
| **3** | 4 | 1 | 1 | 1 | 0 | S | 2 | 35.0 | 1 |
| **4** | 5 | 0 | 3 | 0 | 0 | S | 1 | 35.0 | 1 |

In [29]:
```
#Renaming "age" and "gender" columns
titanic.rename(columns={'age':'Age'}, inplace=True)
titanic.head()
```

Out[29]:

| | PassengerId | Survived | Pclass | SibSp | Parch | Embarked | Gender | Age |
|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 1 | 0 | S | 1 | 22.0 |
| **1** | 2 | 1 | 1 | 1 | 0 | C | 2 | 38.0 |
| **2** | 3 | 1 | 3 | 0 | 0 | S | 2 | 26.0 |
| **3** | 4 | 1 | 1 | 1 | 0 | S | 2 | 35.0 |
| **4** | 5 | 0 | 3 | 0 | 0 | S | 1 | 35.0 |

In [30]:
```
 titanic.rename(columns={'Gender':'Sex'}, inplace=True)
titanic.head()
```

Out[30]:

| | PassengerId | Survived | Pclass | SibSp | Parch | Embarked | Sex | Age |
|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 1 | 0 | S | 1 | 22.0 |
| **1** | 2 | 1 | 1 | 1 | 0 | C | 2 | 38.0 |
| **2** | 3 | 1 | 3 | 0 | 0 | S | 2 | 26.0 |
| **3** | 4 | 1 | 1 | 1 | 0 | S | 2 | 35.0 |
| **4** | 5 | 0 | 3 | 0 | 0 | S | 1 | 35.0 |

In [31]:
```
def getEmb(str):
    if str=="S":
        return 1
    elif str=='Q':
        return 2
    else:
        return 3
titanic["Embark"]=titanic["Embarked"].apply(getEmb)
titanic.head()
```

Out[31]:

| | PassengerId | Survived | Pclass | SibSp | Parch | Embarked | Sex | Age | Embark |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 1 | 0 | S | 1 | 22.0 | 1 |
| **1** | 2 | 1 | 1 | 1 | 0 | C | 2 | 38.0 | 3 |
| **2** | 3 | 1 | 3 | 0 | 0 | S | 2 | 26.0 | 1 |
| **3** | 4 | 1 | 1 | 1 | 0 | S | 2 | 35.0 | 1 |
| **4** | 5 | 0 | 3 | 0 | 0 | S | 1 | 35.0 | 1 |

In [35]:
```
del titanic['Embarked']
titanic.rename(columns={'Embark':'Embarked'}, inplace=True)
```

```
titanic.head()
```

Out[35]:

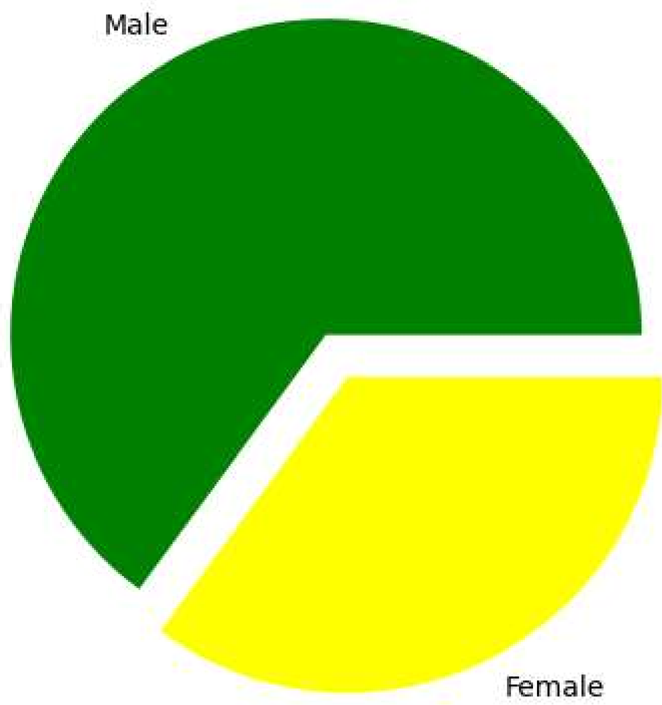
| | PassengerId | Survived | Pclass | SibSp | Parch | Sex | Age | Embarked |
|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 1 | 0 | 1 | 22.0 | 1 |
| **1** | 2 | 1 | 1 | 1 | 0 | 2 | 38.0 | 3 |
| **2** | 3 | 1 | 3 | 0 | 0 | 2 | 26.0 | 1 |
| **3** | 4 | 1 | 1 | 1 | 0 | 2 | 35.0 | 1 |
| **4** | 5 | 0 | 3 | 0 | 0 | 1 | 35.0 | 1 |

In [36]:

```
#Drawing a pie chart for number of males and females aboard
import matplotlib.pyplot as plt
from matplotlib import style

males = (titanic['Sex'] == 1).sum()
#Summing up all the values of column gender with a
#condition for male and similary for females
females = (titanic['Sex'] == 2).sum()
print(males)
print(females)
p = [males, females]
plt.pie(p,    #giving array
       labels = ['Male', 'Female'], #Correspndingly giving labels
       colors = ['green', 'yellow'],   # Corresponding colors
       explode = (0.15, 0),    #How much the gap should me there between the pies
       startangle = 0)  #what start angle should be given
plt.axis('equal')
plt.show()
```

```
577
312
```




In [37]:
```
# More Precise Pie Chart
MaleS=titanic[titanic.Sex==1][titanic.Survived==1].shape[0]
print(MaleS)
MaleN=titanic[titanic.Sex==1][titanic.Survived==0].shape[0]
print(MaleN)
FemaleS=titanic[titanic.Sex==2][titanic.Survived==1].shape[0]
print(FemaleS)
FemaleN=titanic[titanic.Sex==2][titanic.Survived==0].shape[0]
print(FemaleN)
```

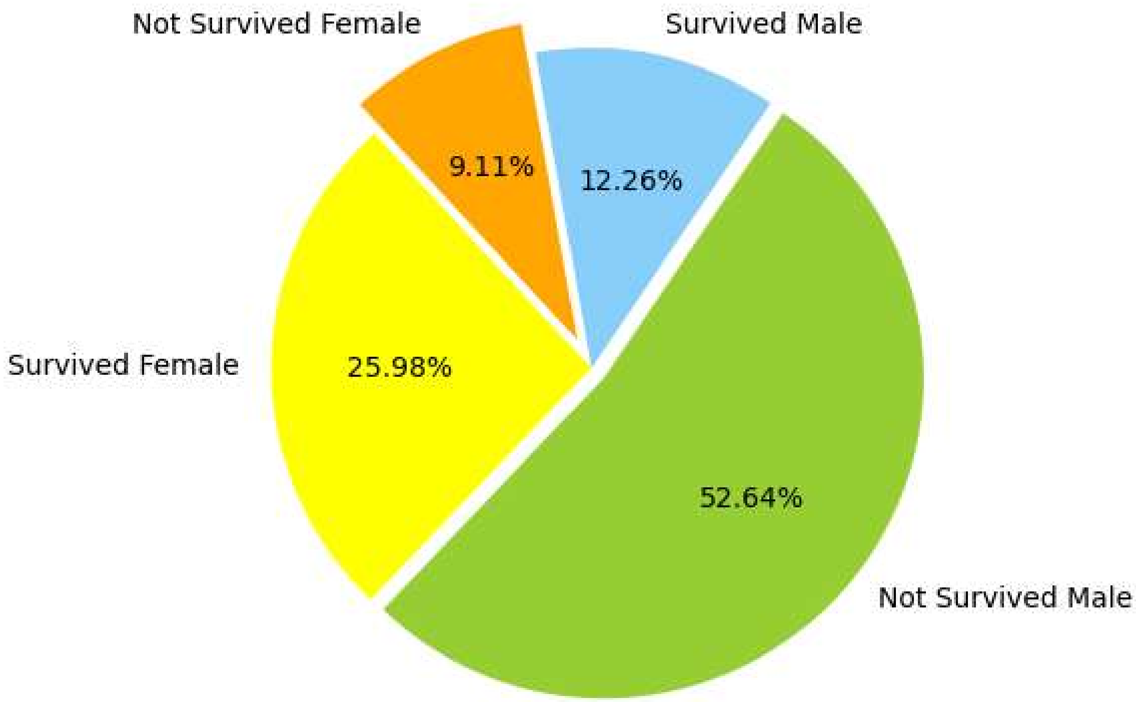
```
109
468
231
81
```

```
C:\Users\LIKHITH\AppData\Local\Temp\ipykernel_18756\3105620411.py:2: UserWarning: Bo
olean Series key will be reindexed to match DataFrame index.
  MaleS=titanic[titanic.Sex==1][titanic.Survived==1].shape[0]
C:\Users\LIKHITH\AppData\Local\Temp\ipykernel_18756\3105620411.py:4: UserWarning: Bo
olean Series key will be reindexed to match DataFrame index.
  MaleN=titanic[titanic.Sex==1][titanic.Survived==0].shape[0]
C:\Users\LIKHITH\AppData\Local\Temp\ipykernel_18756\3105620411.py:6: UserWarning: Bo
olean Series key will be reindexed to match DataFrame index.
  FemaleS=titanic[titanic.Sex==2][titanic.Survived==1].shape[0]
C:\Users\LIKHITH\AppData\Local\Temp\ipykernel_18756\3105620411.py:8: UserWarning: Bo
olean Series key will be reindexed to match DataFrame index.
  FemaleN=titanic[titanic.Sex==2][titanic.Survived==0].shape[0]
```

In [38]:
```
plt.show()
```

In [39]:
```
chart=[MaleS,MaleN,FemaleS,FemaleN]
colors=['lightskyblue','yellowgreen','Yellow','Orange']
labels=["Survived Male","Not Survived Male","Survived Female","Not Survived Female"
```

```
explode=[0,0.05,0,0.1]
plt.pie(chart,labels=labels,colors=colors,explode=explode,startangle=100,counterclo
plt.axis("equal")
plt.show()
```




```
In [ ]:
```